In OpenStack, the **hypervisor**—also known as the **Virtual Machine Monitor (VMM)**—is the core of the compute node, providing the manageability functions that allow virtual machines to access the underlying physical hardware 1. Choosing and managing hypervisors involves a combination of architectural planning and automated selection via the Nova service.

Types of Hypervisors Supported

OpenStack is designed to offer a wide range of VMMs to suit different infrastructure needs. Supported hypervisors include:

- **KVM (Kernel-based Virtual Machine):** A popular open-source option known for its efficiency and security 1, 2.
- **VMware ESXi:** An enterprise-grade solution offering advanced features, though it requires commercial licensing 1, 3.
- **Hyper-V:** Microsoft's solution, favored for its tight integration with Windows environments 1, 3.
- **Others:** QEMU, Xen, UML (User Mode Linux), LXC (Linux Containers), bare metal, and Docker containers 1.

Selection Criteria

When deciding on a hypervisor, administrators must evaluate several key factors:

- **Performance Requirements:** The hypervisor's capabilities must align with the resource demands of the specific workloads 3.
- **Cost:** Organizations must weigh the benefits of commercial enterprise-grade features against the licensing costs of solutions like VMware versus free open-source alternatives like KVM 3.
- **Team Skillset:** Choosing a hypervisor that the administrative team is already familiar with ensures easier management and lower operational overhead 3.
- **Feature Parity:** Not all hypervisors support the same OpenStack features. Administrators should consult the **Hypervisor Support Matrix** to ensure their chosen VMM supports the necessary functionalities 4.

Performance Considerations

- **Overcommitment:** Administrators can allocate more CPU and memory than is physically available (overcommitment) to improve utilization 5. However, this carries the risk of **performance degradation** if those resources are heavily utilized simultaneously 5.
- **Sizing:** Correctly sizing compute nodes is essential to avoid waste 6. Technologies like **hyper-threading** are highly recommended, as they can effectively double the number of available cores per node 7.
- **External Storage:** Hosting instance disks on external shared storage can improve recovery times during node failure but may introduce performance issues due to **network latency** or heavy I/O from neighboring VMs 8.

Compatibility and Image Requirements

The image service (**Glance**) must store templates in formats compatible with the target hypervisor 9. Supported formats include:

- **RAW and QCOW2** (typically for KVM/QEMU) 9.
- **VMDK** (for VMware) and **OVA** 9.
- **Metadata:** Users can associate metadata with images (e.g., OS version, architecture) to help the system identify compatible compute nodes during deployment 10, 11.

The Selection Process: Nova-Scheduler

The actual placement of a VM on a specific hypervisor/compute node is performed by the **nova-scheduler** 12. It follows a three-step logic:

1. **Filtering:** It eliminates nodes that are unsuitable due to maintenance, lack of resources, or missing hardware features like **GPUs or accelerators** 13, 14.
2. **Weighting:** Remaining nodes are ranked based on suitability, such as favoring those with low memory utilization or proximity to other VMs for better network performance 14, 15.
3. **Selection:** The node with the highest weight is selected as the optimal host 15.

Exam-Ready Summary

- **Definition:** The hypervisor/VMM is the "heart" of the compute node, enabling VMs to interact with hardware 1.
- **Key Comparison: KVM** (Efficient/Open-source) vs. **VMware** (Enterprise features/Commercial) vs. **Hyper-V** (Windows integration) 2, 3.
- **Selection Process:** Selection is handled by **Nova-scheduler** via **Filtering** (eliminating the unfit) and **Weighting** (ranking the fit) 14, 15.
- **Performance Tip:** Use **hyper-threading** to maximize core counts and carefully manage **overcommitment** to balance utilization and performance 5, 7.
- **Redundancy:** Use **Availability Zones** to group compute nodes into different fault domains to ensure service continuity during outages 7, 16.